

```
import pandas as pd
from sklearn.preprocessing import MinMaxScaler, LabelEncoder
from sklearn.decomposition import PCA
```

```
EmpleadosAttrition = pd.read_csv('empleadosRETO.csv')
EmpleadosAttrition.head()
```

	Age	BusinessTravel	Department	DistanceFromHome	Education	EducationField
0	50	Travel_Rarely	Research & Development	1 km	2	Medical
1	36	Travel_Rarely	Research & Development	6 km	2	Medical
2	21	Travel_Rarely	Sales	7 km	1	Marketing
3	52	Travel_Rarely	Research & Development	7 km	4	Life Sciences
4	33	Travel_Rarely	Research & Development	15 km	1	Medical

5 rows × 30 columns

```
# Eliminar columnas que no aportan al análisis
EmpleadosAttrition = EmpleadosAttrition.drop(columns=['EmployeeCount', 'Employee
```

```
# Calcular los años que el empleado lleva en la compañía hasta 2018
EmpleadosAttrition['Year'] = EmpleadosAttrition['HiringDate'].astype(str).str[-4:]
EmpleadosAttrition['YearsAtCompany'] = 2018 - EmpleadosAttrition['Year']
```

```
# Convertir DistanceFromHome a número entero
EmpleadosAttrition = EmpleadosAttrition.rename(columns={'DistanceFromHome': 'DistanceFromHome_k'})
EmpleadosAttrition['DistanceFromHome'] = EmpleadosAttrition['DistanceFromHome_k'] * 1000
```

```
EmpleadosAttrition = EmpleadosAttrition.drop(columns=['Year', 'HiringDate', 'DistanceFromHome_k'])
```

```
# Tabla informativa de sueldo promedio por departamento
SueldoPromedioDepto = EmpleadosAttrition.groupby('Department')['MonthlyIncome'].mean()
SueldoPromedioDepto = SueldoPromedioDepto.rename(columns={'MonthlyIncome': 'SueldoPromedioDepto'})
```

	Department	SueldoPromedio	
0	Human Resources	6239.888889	
1	Research & Development	6804.149813	
2	Sales	7188.250000	

Next steps:

[Generate code with SueldoPromedioDepto](#)[New interactive sheet](#)

```
# Escalar MonthlyIncome entre 0 y 1
escalador = MinMaxScaler()
EmpleadosAttrition['MonthlyIncome'] = escalador.fit_transform(EmpleadosAttritic
```

```
# Convertir variables categóricas a numéricas
categoricas = EmpleadosAttrition.select_dtypes(include='object').columns
codificador = LabelEncoder()

for columna in categoricas:
    EmpleadosAttrition[columna] = codificador.fit_transform(EmpleadosAttrition[
```

```
# Correlación lineal de cada variable con Attrition
CorrelacionAttrition = EmpleadosAttrition.corr()['Attrition'].sort_values(ascr
CorrelacionAttrition
```

	Attrition
Attrition	1.000000
OverTime	0.324777
MaritalStatus	0.187283
JobRole	0.078684
BusinessTravel	0.060677
Department	0.054236
DistanceFromHome	0.052732
EducationField	0.051184
PerformanceRating	-0.006471
NumCompaniesWorked	-0.009082
WorkLifeBalance	-0.021723
Gender	-0.028839
RelationshipSatisfaction	-0.030945
Education	-0.055531
PercentSalaryHike	-0.060880
YearsSinceLastPromotion	-0.069000
TrainingTimesLastYear	-0.070884
EnvironmentSatisfaction	-0.124327
JobSatisfaction	-0.164957
JobInvolvement	-0.166785
YearsAtCompany	-0.176001
MonthlyIncome	-0.194936
YearsInCurrentRole	-0.203918
Age	-0.212121
TotalWorkingYears	-0.213329
JobLevel	-0.214266

dtype: float64

```
# Seleccionar variables con correlación mayor o igual a 0.1
variables_seleccionadas = CorrelacionAttrition[abs(CorrelacionAttrition) >= 0
```

```

if 'Attrition' not in variables_seleccionadas:
    variables_seleccionadas.append('Attrition')

EmpleadosAttritionFinal = EmpleadosAttrition[variables_seleccionadas].copy()
EmpleadosAttritionFinal.head()

```

	Attrition	OverTime	MaritalStatus	EnvironmentSatisfaction	JobSatisfaction
0	0	0	0	4	4
1	0	0	0	2	2
2	1	0	2	2	2
3	0	0	2	2	2
4	1	1	1	2	3

Next steps:

[Generate code with EmpleadosAttritionFinal](#)[New interactive sheet](#)

```

# Crear componentes principales y agregar los necesarios para explicar al menos
pca = PCA()
EmpleadosAttritionPCA = pca.fit_transform(EmpleadosAttritionFinal)

varianza_acumulada = pca.explained_variance_ratio_.cumsum()
num_componentes = (varianza_acumulada < 0.80).sum() + 1

for i in range(num_componentes):
    EmpleadosAttritionFinal = EmpleadosAttritionFinal.assign(**{f'C{i}': Emplea

EmpleadosAttritionFinal.head()

```

	Attrition	OverTime	MaritalStatus	EnvironmentSatisfaction	JobSatisfaction
0	0	0	0	4	4
1	0	0	0	2	2
2	1	0	2	2	2
3	0	0	2	2	2
4	1	1	1	2	3